7. Wealth Inequality

Adv. Macro: Heterogeneous Agent Models

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Roadmap for the rest of the class

- We are done with the "computational" part of the class!
- We can switch to "economics" :-)

Now:

7. Today: Wealth Inequality
8. Next week: Secular Stagnation

After this, fiscal and monetary policy in HANK, SAM, I-HANK...

Introduction

Wealth inequality

- **Goal for today:** understand wealth inequality through the lens of heterogeneous agent models
- **Central economic questions:**
  1. Why are some people so rich?
  2. What explains the rise in wealth inequality since 1980?
  3. What can taxes do about it?
- **Plan for today:**
  1. Facts on wealth inequality
  2. Pareto tails, random growth, and why the buffer-stock model fails
  3. Mechanisms that fix it: Gaillard et al. (2024), Guvenen et al. (2023)
  4. Drivers of the rise: Hubmer, Krusell and Smith (2021), Moll, Rachel and Restrepo (2022)

Data

Earnings and wealth inequality

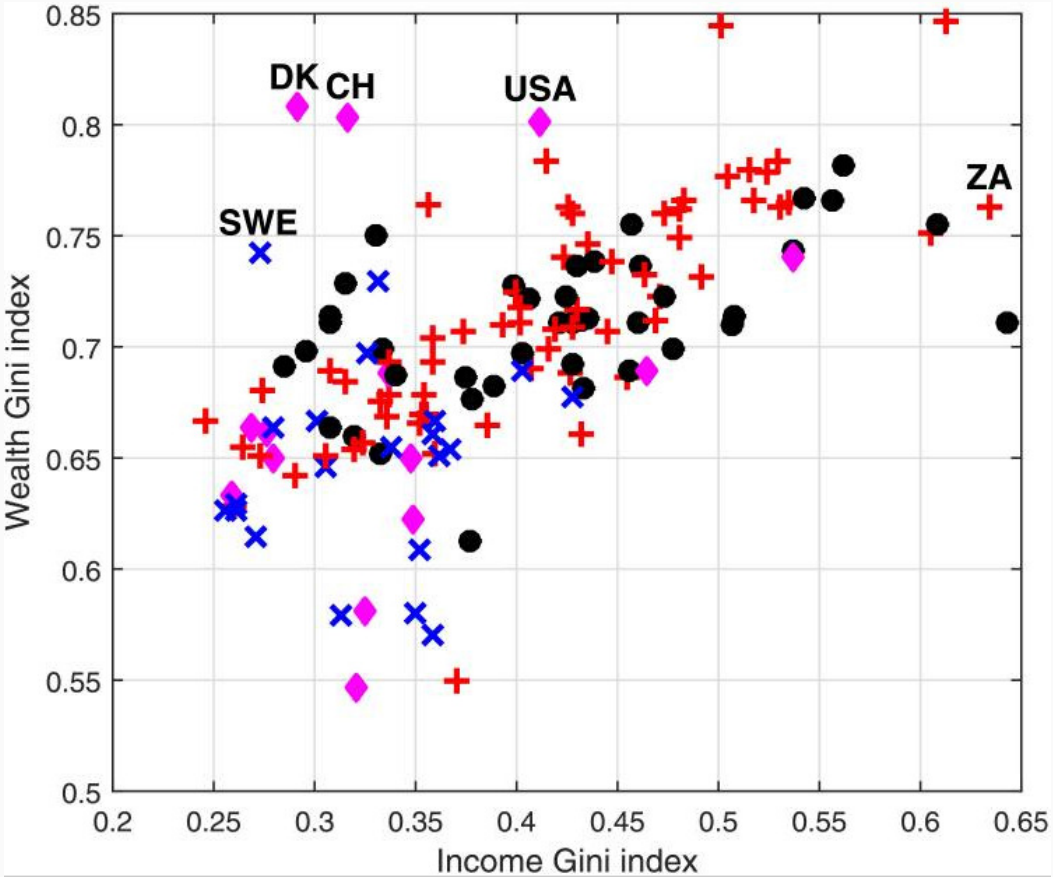
- US data on the distribution of earnings and wealth (SCF, 1989)

	Top 1%	Top 5%	Top 20%	Top 40%	Zero or negative
Wealth	29	53	80	93	6
Earnings	6	19	48	72	8

- Wealth is far more concentrated than earnings
- Skewed distributions with thick upper tails

Wealth more concentrated than earnings

Not only in the US, but in almost all countries, Denmark included



→ Denmark: very unequal wealth despite equal incomes

Wealth has a Pareto tail

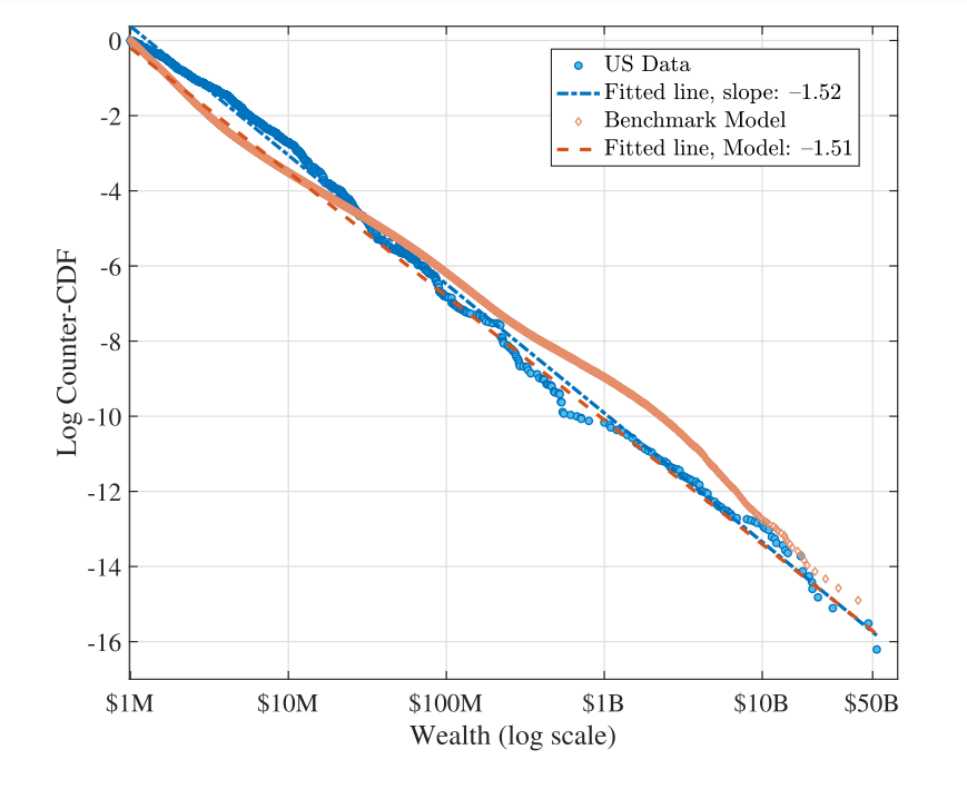


Figure 1: US wealth above \$1M, Guvenen et al. (2023)

$\log P(a > x)$ is linear in $\log x$, with slope ≈ -1.5, from \$1M to \$50B

Ranking of Pareto tails

Empirical ranking of Pareto coefficients (US, PSID):

$$\zeta_{ra} < \zeta_a < \zeta_y < \zeta_c$$

capital income $>$ wealth $>$ labor income $>$ consumption in concentration

Table 1. Top consumption, income, and wealth ineq

Data	Variable	Best fit Pareto^b			
		$\hat{\underline{x}}^{OLS}$	$\hat{\zeta}^{OLS}$	$\hat{\underline{x}}^{MLE}$	$\hat{\zeta}^{MLE}$
PSID	Capital income	0.96	**1.22**	0.96	**1.21**
		(0.02)	(0.15)	(0.02)	(0.14)
	Wealth	0.93	**1.48**	0.92	**1.47**
		(0.03)	(0.09)	(0.03)	(0.09)
	Labor income	0.88	**2.42**	0.89	**2.50**
		(0.04)	(0.15)	(0.04)	(0.13)
	Consumption	0.89	**3.11**	0.90	**3.13**
		(0.04)	(0.28)	(0.04)	(0.20)
	Food consumption	0.93	**4.40**	0.93	**4.26**
		(0.05)	(0.33)	(0.05)	(0.43)

Figure 2: Table 1 from Gaillard et al. (2024)

Adjusted for under-reporting: 1.20, 1.38, 2.25, 3.06

Richer households hold riskier assets

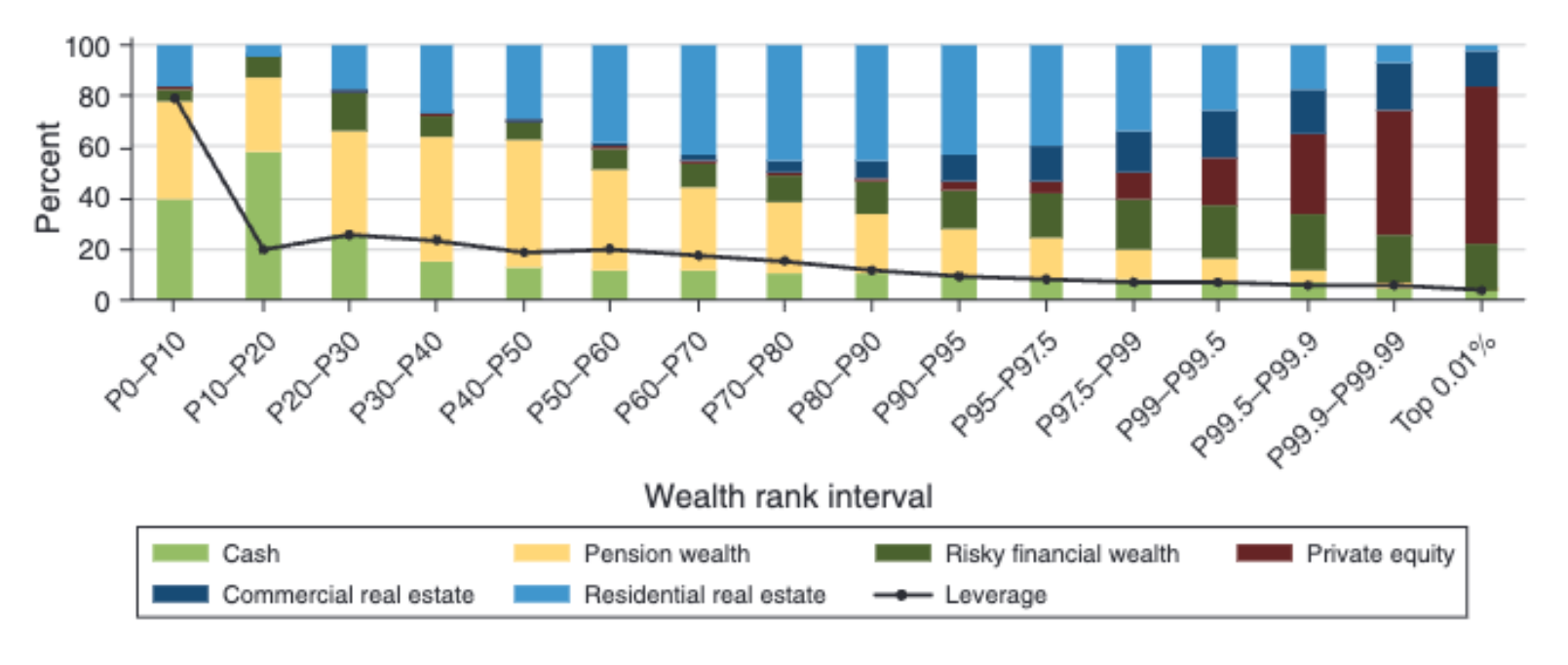


Figure 3: Figure 2 from Bach, Calvet and Sodini (2020), Swedish registers

Bottom: cash and pensions. Middle: housing. Top: equity and private equity.

Richer households have higher returns

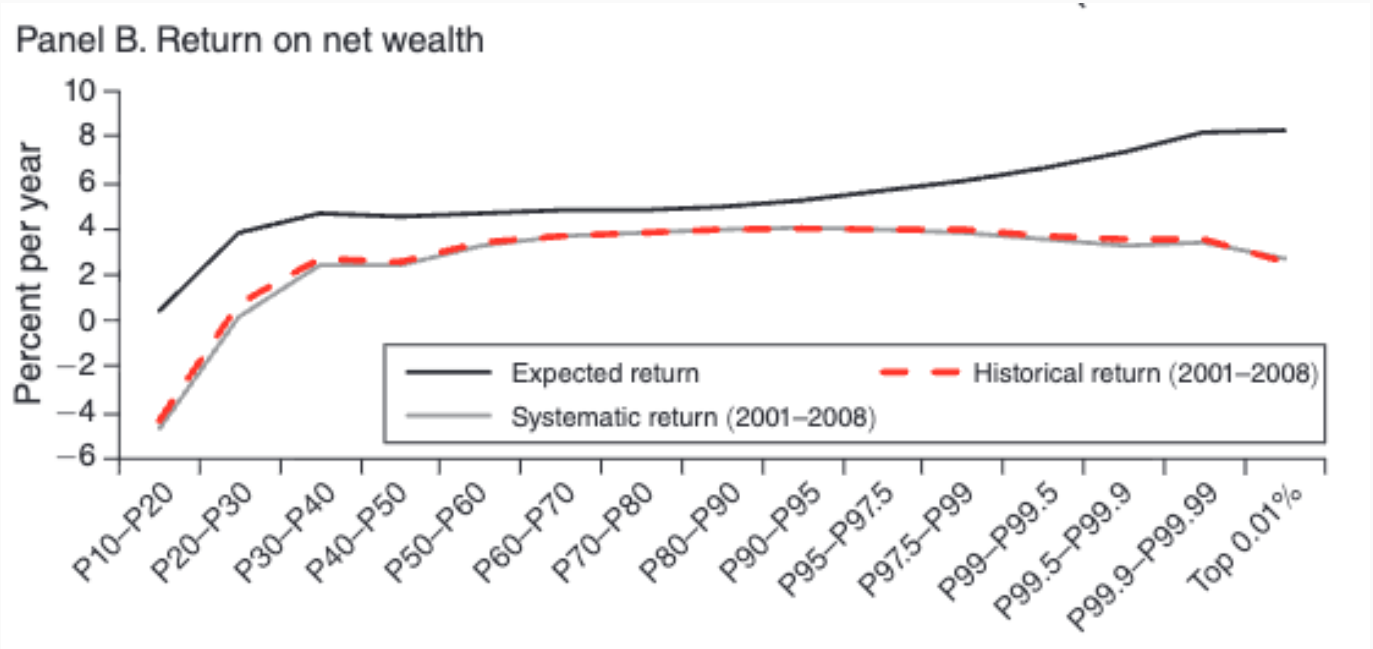


Figure 4: Figure 3 from Bach, Calvet and Sodini (2020)

- Expected returns rise with net worth. Returns also differ within asset classes, are persistent and correlated across generations (Fagereng, Guiso, Malacrino and Pistaferri, 2020)
- Still a debate: higher risk or higher skill?

Why do the rich earn more: risk or skill?

Risk: Bach, Calvet and Sodini (2020, AER), Sweden

- Expected returns rise with net worth, driven mainly by *systematic risk*: the rich hold more equity and private equity
- Top 0.01%: excess return $\approx$ the equity premium
- Idiosyncratic risk is transitory, but creates large long-run dispersion at the top

Skill: Fagereng, Guiso, Malacrino and Pistaferri (2020, Ecta), Norway

- Returns differ even *within* narrow asset classes
- p10 to p90 of net worth: +18 pp (+10 pp after tax)
- Persistent: partly risk and scale, but also sophistication, information and entrepreneurial talent

- Why it matters: with skill, high returns mark productive households, a persistent type; a wealth tax then shifts the burden away from them (Guvenen et al., 2023)

The rich save more, because of capital gains

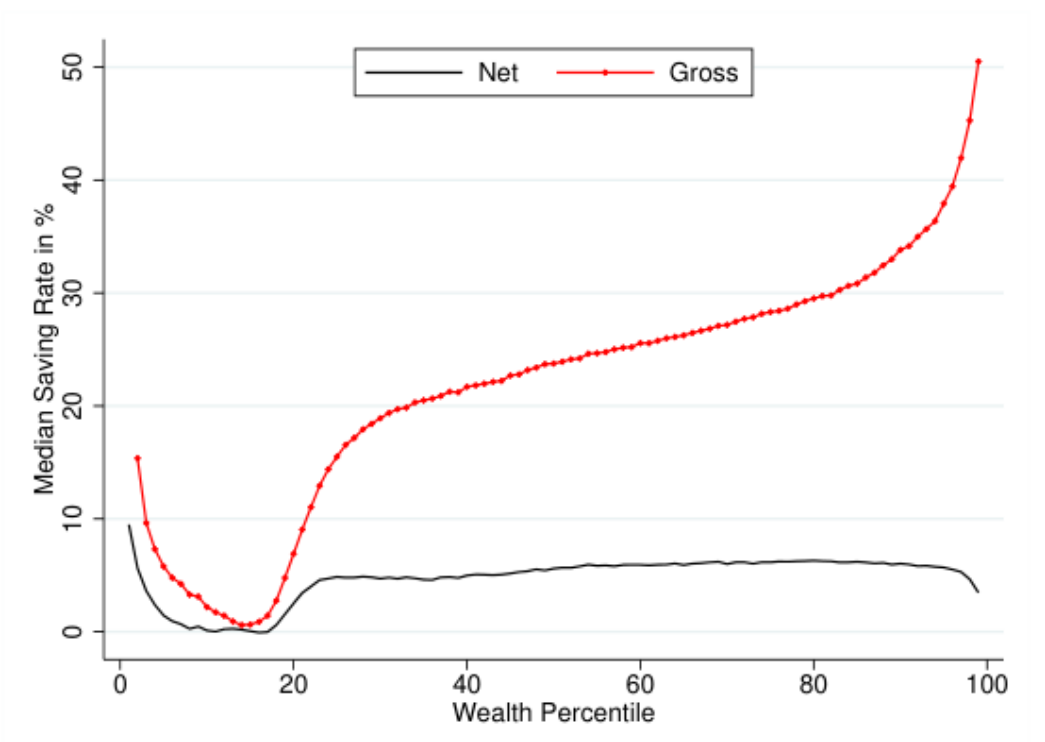


Figure 5: Figure 1 from Fagereng, Holm, Moll and Natvik (2019), Norway

Gross saving rates rise to 50% at the top, net saving rates are flat: the rich save by not consuming their capital gains

How do people become rich?

- Hubmer, Halvorsen, Salgado and Ozkan (forthcoming, Econometrica): Norwegian registers, 1993 to 2019
- Budget constraint, with s_{it} the gross saving rate out of total income:

$$a_{it} = a_{it-1} + (r_{it} a_{it-1} + L_{it} + H_{it}) s_{it}$$

  returns r (incl. capital gains), labor income L, inheritances and gifts H
- Decomposition: replace one component by the average of the middle 50% of the same age, simulate wealth, average over all orderings (Shapley-Owen)
- Wealth at age 20 counts as inheritance

The top 0.1%: saving, inheritances and returns

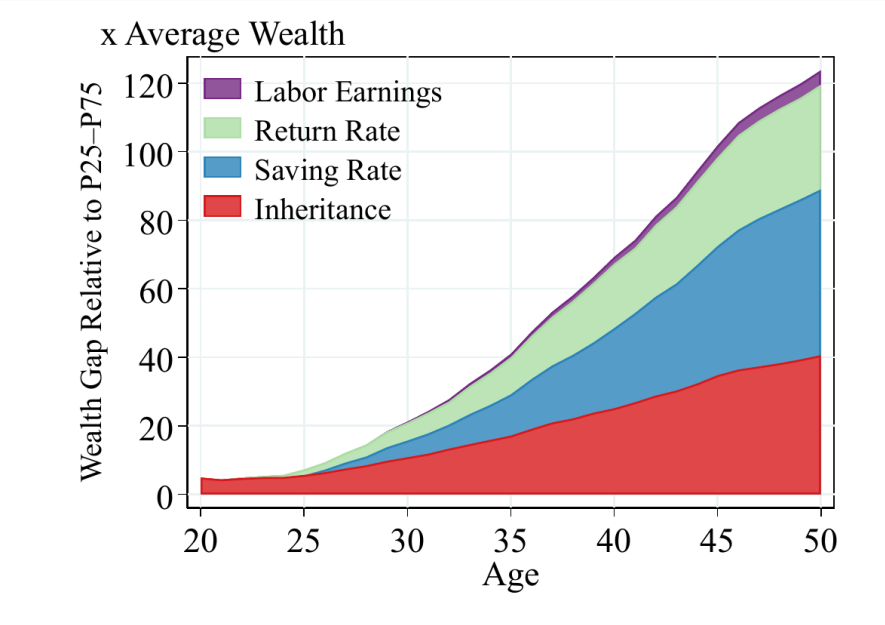
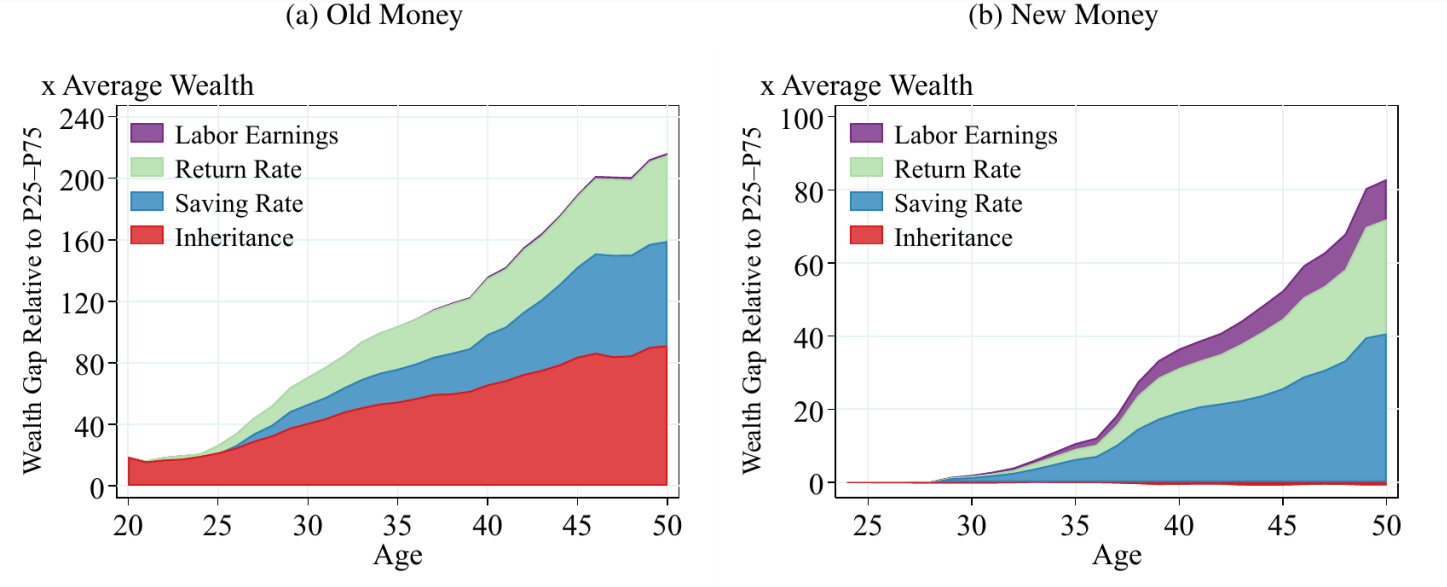


Figure 11 from Hubmer et al. (forthcoming)

At age 50, the top 0.1% hold 124 $\times$ average wealth. Share of their gap to the middle 50%:

- saving rates: 39%
- inheritances: 33%
- returns: 25%
- labor income: 3%

Old Money and New Money

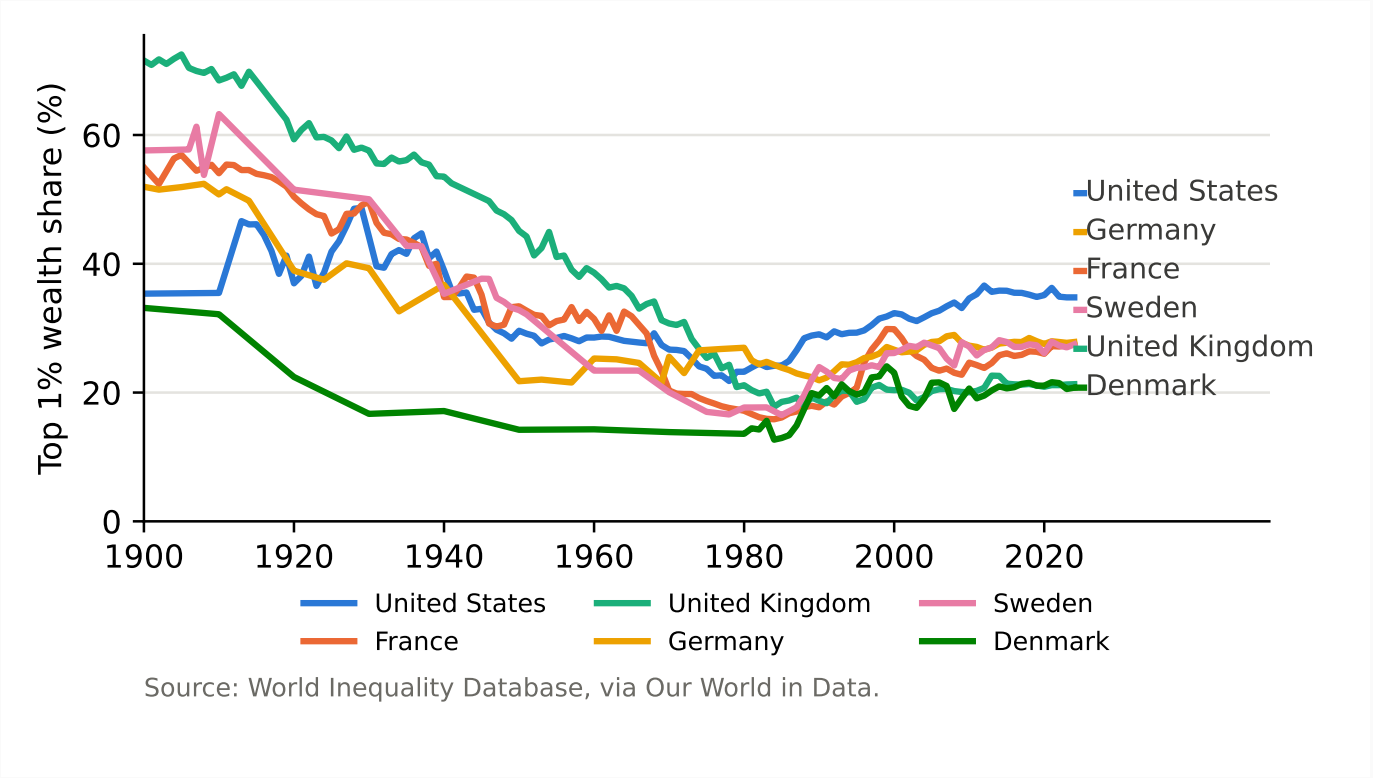


- **Old Money** (a quarter, already top 0.1% in their mid-20s): inheritances 42%, saving 31%, returns 26%, labor 1%
- **New Money** (a quarter, start with negative wealth): saving 50%, returns 38%, labor 13%, inheritances −1%

How do people become rich? Takeaways

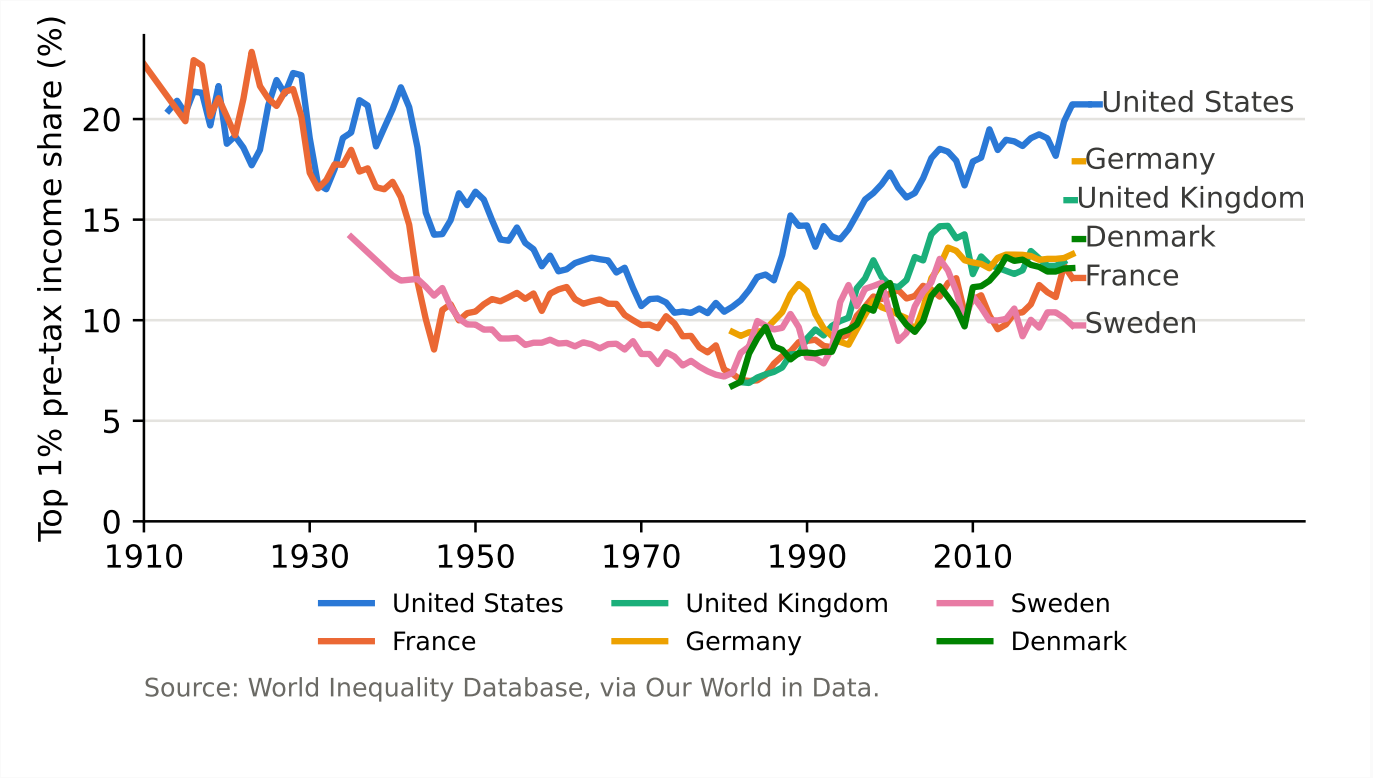
- Saving rates are the largest channel (saving includes retained business profits)
- Family money matters, once transfers early in life are counted
- Self-made New Money: high saving, high returns from private businesses, and high labor income
- Labor income matters little, except for New Money

Top wealth shares: US and Europe



- U-shape: large fall until 1980, rise since, much stronger in the US
- Europe was more unequal than the US before 1914

Top income shares: US and Europe



- US top 1%: from about 10% around 1980 to about 20% today
- Europe: a smaller rise, to 10–13%

Income growth by percentile in the US

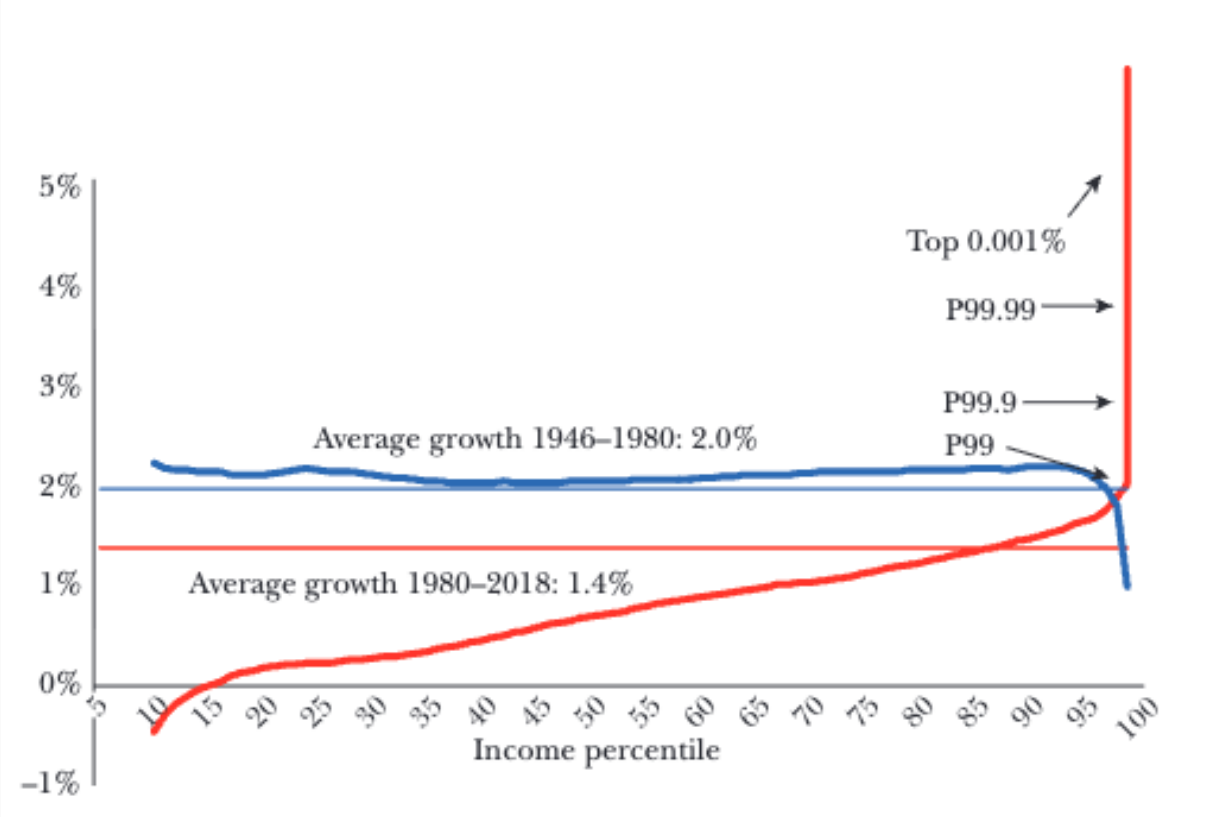


Figure 6: Figure 4 from Saez and Zucman (2020)

Average tax rates by income groups (US)

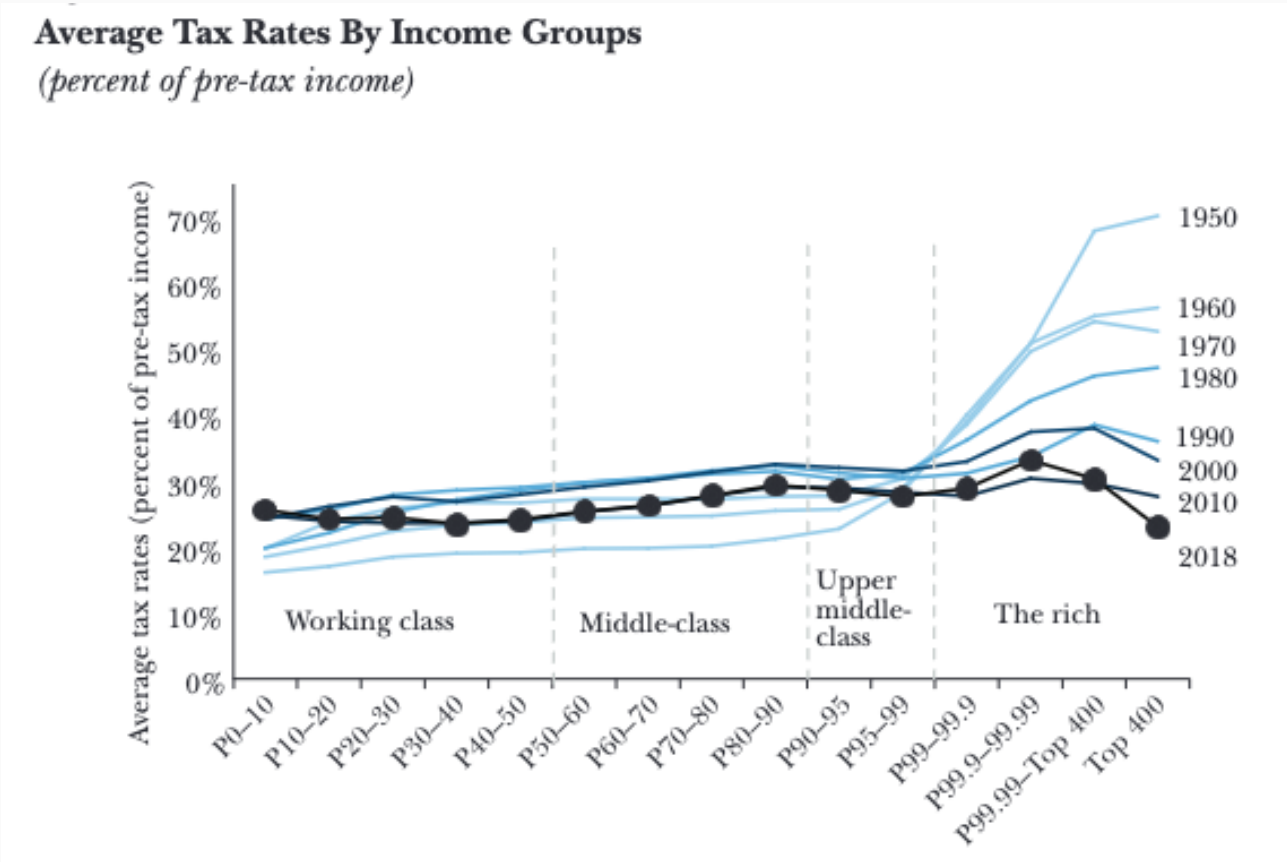


Figure 7: Figure 5 from Saez and Zucman (2020)

Top 400: about 70% in 1950, about 23% in 2018

Taking stock

- Wealth is more concentrated than income, with a Pareto tail
- Ranking of tails: capital income $>$ wealth $>$ labor income $>$ consumption
- Returns are heterogeneous, and higher for richer households
- The rich save more, mostly through capital gains
- Wealth and income inequality have increased since 1980
- Taxes became less progressive (at least in the US)

Pareto tails

Pareto distribution

- Wealth a is Pareto with minimum a_0 and tail index ζ if

$$P\left(a > x\right) = \left(\frac{x}{a_0}\right)^{-\zeta}, \quad x \geq a_0$$

- Pareto tail: $P(a > x) \approx Cx^{-\zeta}$ for large x. Lower ζ = fatter tail

Properties of the Pareto distribution

- **P1. Moments:** $E\left[a^\theta\right] = \frac{\zeta a_0^\theta}{\zeta - \theta}$ if $\theta < \zeta$, ∞ otherwise (mean wealth is finite only if $\zeta > 1$)
- **P2. Van der Wijk's law:** above x, a is Pareto with minimum x, so

$$E\left[a \mid a > x\right] = \frac{\zeta}{\zeta - 1} x$$

- **P3. Top shares:** the top fraction q holds

$$S(q) = q^{1-1/\zeta}$$

  - $\zeta = 1.5$: top 1% holds $0.01^{1/3} \approx 22\%$
- **Power rule:** if $X = a^b$ with $b > 0$, then $\zeta_X = \zeta_a / b$
  - proof: $P(X > x) = P(a > x^{1/b}) \propto x^{-\zeta_a / b}$
  - concave functions of wealth ($b < 1$) have thinner tails, convex ones ($b > 1$) fatter tails

Wealth as a random growth process

- Budget constraint: $a_{it} = (1 + r_{it})a_{it-1} + wz_{it} - c_{it}$
- Let households save a share s_{it} of cash-on-hand:

$$a_{it} = G_{it}\, a_{it-1} + y_{it}, \qquad G_{it} = s_{it}(1 + r_{it}), \qquad y_{it} = s_{it}\, wz_{it}$$

- G_{it}: growth factor, scales with wealth
- y_{it}: saved labor income, does not scale with wealth
- Assume $(G_{it}, y_{it}) \geq 0$ i.i.d. over time and households, independent of a_{it-1}
- Two routes to a Pareto tail of wealth:
  1. Pareto labor income: y_{it} has a Pareto tail
  2. Random growth: G_{it} random

Route 1: Pareto labor income

- $G \in (0,1)$ constant, y_{it} i.i.d. with a Pareto tail, $P(y > x) \approx Cx^{-\zeta_y}$
- Wealth is a weighted sum of past incomes:

$$a_{it} = \sum_{k \geq 0} G^k y_{i,t-k}$$

- Large when one term is large:

$$P(a_{it} > x) \approx \sum_k C \left(\frac{x}{G^k} \right)^{-\zeta_y} \propto x^{-\zeta_y}$$

- Wealth inherits the labor income tail: $\zeta_a = \zeta_y$
- Data: $\zeta_a = 1.38 < \zeta_y = 2.25 \Rightarrow$ labor income alone cannot explain the wealth tail, we need random growth (Stachurski and Toda, 2019)

Route 2: random growth

- G_{it} i.i.d., random through returns or saving rates, with
  - $P(G > 1) > 0$: some households grow by luck
  - $E[\ln G] < 0$: something stops them (they shrink on average, or die)
  - y_{it} thin-tailed (e.g. lognormal), positive with positive probability
- One can show (Kesten, 1973) that wealth has a Pareto tail, $P(a > x) \approx Cx^{-\zeta}$, where $\zeta > 0$ solves

$$E\left[G^{\zeta}\right] = 1$$

- Lognormal G, $\ln G \sim \mathcal{N}(\mu_G, \sigma_G^2)$ with $\mu_G < 0$:

$$\zeta = -\frac{2\mu_G}{\sigma_G^2}$$

  - $\zeta =$ pull back down $(-\mu_G)$ relative to luck $(\sigma_G^2/2)$: more luck or less shrinking gives a fatter tail

Proof

Constant $\bar{G}$ + reset model

- Survivors' wealth grows by a constant $\bar{G} > 1$; each period households die with i.i.d. probability ξ and are replaced by newborns with wealth a_0
- Exact solution by age: wealth at age s is $a_0 \bar{G}^s$, and the share aged s or more is $(1-\xi)^s$, so

$$P(a > x) = P\left(\text{age} \geq \frac{\ln(x/a_0)}{\ln \bar{G}}\right) = (1-\xi)^{\ln(x/a_0)/\ln \bar{G}} = \left(\frac{x}{a_0}\right)^{-\zeta}$$

$$\zeta = -\frac{\ln(1-\xi)}{\ln \bar{G}} \approx \frac{\xi}{\ln \bar{G}} \qquad \Leftrightarrow \qquad (1-\xi)\bar{G}^{\zeta} = 1$$

- Fatter tail when the rich grow faster or live longer. $\xi = 0.02$, $\bar{G} = 1.013$: $\zeta \approx 1.56$

Source: Jones (2015), *Pareto and Piketty*, JEP

Growth, returns and taxes

- Add economic growth: wages and average wealth grow at rate g, $w_t = (1+g)^t w_0$. Also add risky returns (mean $\bar{r}$, std. σ_r), consumption share of the rich $\bar{c}$, and taxes on capital income τ_k, wealth τ_w, labor income τ_ℓ:

$$a_{it} = (1 + (1-\tau_k) r_{it} - \tau_w)\, a_{it-1} + (1-\tau_\ell) w_t z_{it} - c_{it}$$

- Relative to the average, $\hat{a}_{it} = a_{it}/(1+g)^t$:

$$\hat{a}_{it} = G_{it}\hat{a}_{it-1} + y_{it}, \qquad G_{it} = \frac{(1-\bar{c})\,(1+(1-\tau_k)r_{it} - \tau_w)}{1+g}$$

- Lognormal returns and $\ln(1+x) \approx x$ in $\zeta = -2\mu_G/\sigma_G^2$:

$$\zeta \approx 1 - \frac{(1-\tau_k)\bar{r} - \tau_w - g - \bar{c}}{(1-\tau_k)^2\sigma_r^2/2}$$

Discrete-time version of Moll (2014), *Why Piketty says r−g matters for inequality*

Piketty's $r - g$ and taxes

$$\zeta \approx 1 - \frac{(1-\tau_k)\bar{r} - \tau_w - g - \bar{c}}{(1-\tau_k)^2\sigma_r^2/2}$$

- Top inequality $1/\zeta$ rises with $\bar{r} - g$: Piketty's (2014) $r > g$
- But what matters is $\bar{r} - g - \bar{c}$: the return net of growth and of what the rich consume
- Return risk is essential: without it, no Pareto tail. $r > g$ is neither necessary nor sufficient
- $\zeta > 1$ needs $(1-\tau_k)\bar{r} - \tau_w - \bar{c} < g$: on average the rich grow slower than the economy, only the lucky ones climb
- Taxes:
  - wealth tax: lowers mean growth one for one
  - capital income tax: lowers mean growth and risk
  - labor income tax: only enters y, does not change ζ

Taking stock

- Labor income inequality alone cannot generate the relative Pareto tails of wealth / income
- Random growth mechanisms can generate the Pareto tails observed in the data
- The spread $r - g$ increases wealth inequality

But we miss two key features:

- Microfoundations: what determines G, etc?
- General equilibrium: r, g are endogenous to behaviors and the economy

Pareto tails

Failure of the buffer-stock model

Aiyagari model

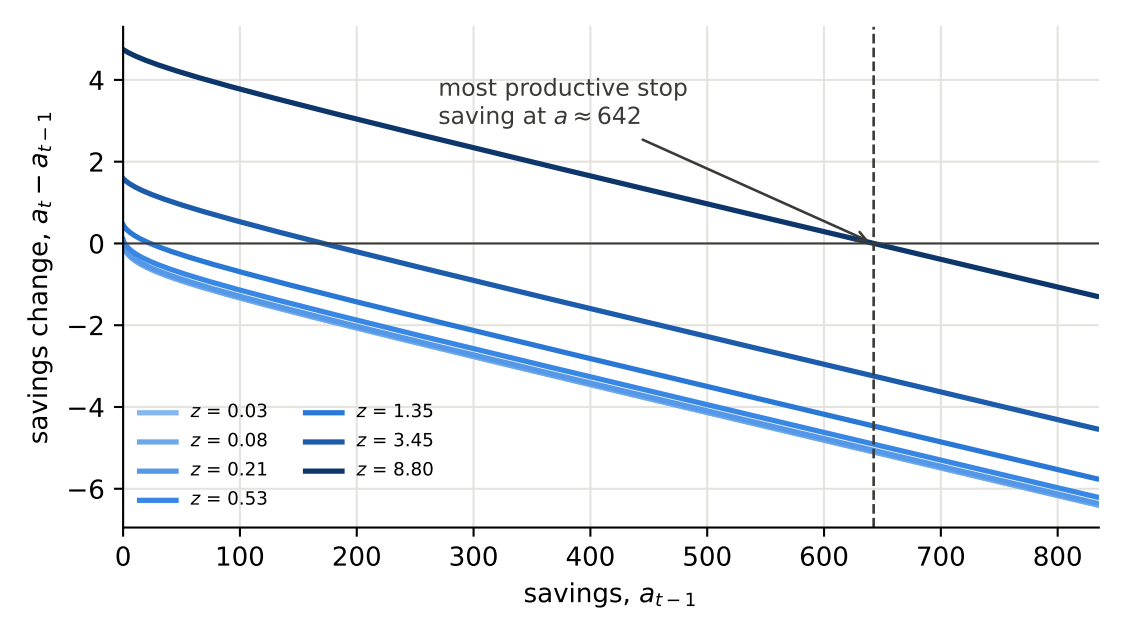
$$\max_{\{c_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \quad \text{s.t.} \quad a_t = (1+r)a_{t-1} + wz_t - c_t, \quad a_t \geq 0$$

with $\ln z_t$ an AR(1) estimated on panel data (Floden and Lindé, 2001)

	Wealth Gini	Top 1%	Top 5%	Top 20%
US data (SCF, 1989)	0.78	29	53	80
Aiyagari baseline	0.38	3.2	12.2	41.0

$\rightarrow$ misses the top by a factor of ten

Policies in the buffer-stock model



- Above some wealth, every household dissaves, even the most productive
- The top type stops at $a \approx 640$, about 40 times average wealth: wealth is bounded, so no Pareto tail

Lecture 4 HANC calibration (quarterly): $\sigma = 1$, $\beta(1 + r) = 0.993$, $\rho_z = 0.9$, $\sigma_\psi = 0.5$

Why the buffer-stock model fails

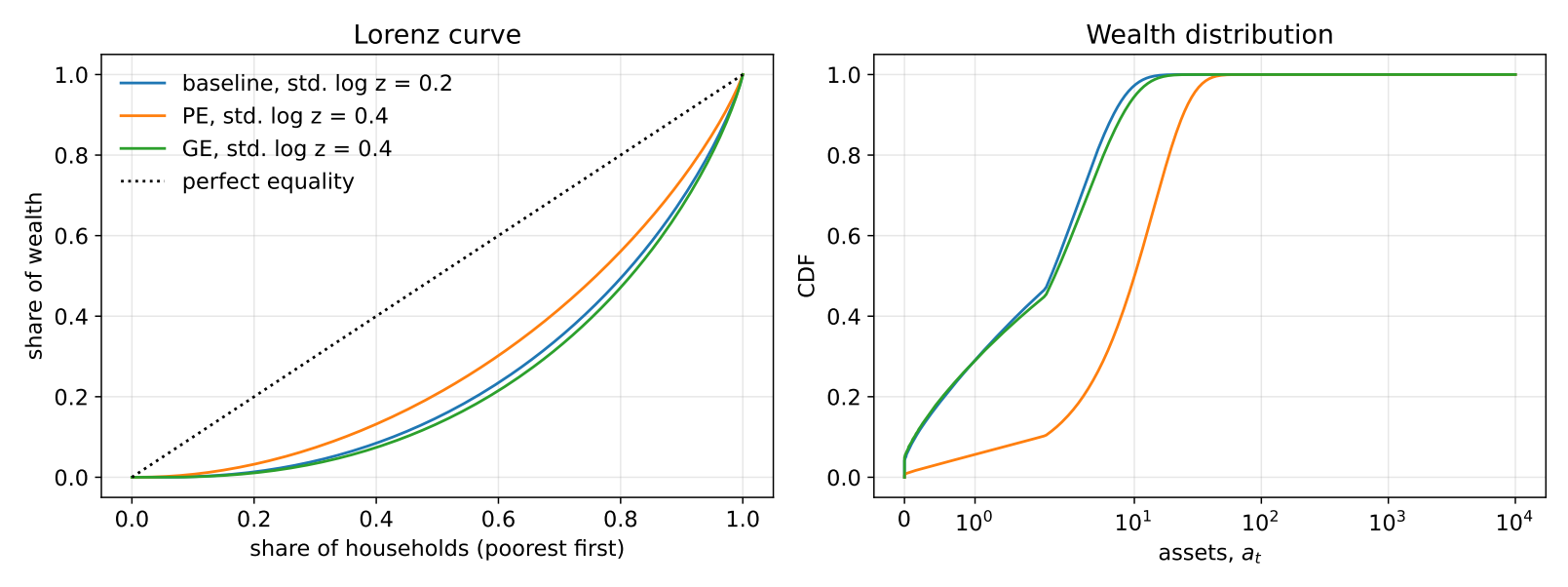
- Precautionary saving: once the buffer-stock target is reached, households dissave (Carroll, 1997)
- At the top, $c = \bar{c}m$ and the Euler equation $c_t^{-\sigma} = \beta(1+r)c_{t+1}^{-\sigma}$ give

$$G_{\text{top}} = (1+r)(1-\bar{c}) = (\beta(1+r))^{1/\sigma} < 1$$

  since $\beta(1+r) < 1$ in equilibrium
- No route 1: an AR(1) gives labor income a thin tail
- No route 2: $G < 1$ and deterministic at the top
- The rich dissave in the model, but save more in the data (Dynan, Skinner and Zeldes, 2004; De Nardi, French and Jones, 2010)

Let's code!

Takeaway: more income risk does not fix the Aiyagari model



- Std. of log income from 0.2 to 0.4. **PE** (r, w fixed): everyone saves more for precaution, the poor most, so wealth inequality *falls* (Gini 0.50 to 0.41)
- **GE**: the lower r undoes most of the extra saving ($K = 3.4$, against 12.0 in PE), and inequality ends slightly *above* the baseline (Gini 0.52)
- Still no Pareto tail: income risk is additive, while a fat tail needs multiplicative risk (route 2) or fat-tailed income (route 1)

Explaining wealth inequality

Four main mechanisms

Each gives some households $G > 1$:

1. **Bequests (or taste for wealth)**
   - Kotlikoff and Summers (1981), Gale and Scholz (1994), De Nardi (2004)
2. **Heterogeneous preferences**
   - Krusell and Smith (1998), Druedahl and Jørgensen (2015)
3. **Entrepreneurship**
   - Quadrini (2000), Cagetti and De Nardi (2006), Guvenen et al. (2023)
4. **Idiosyncratic rates of returns**
   - Benhabib, Bisin and Zhu (2011), Hubmer, Krusell and Smith (2021)

Explaining wealth inequality

$$\max_{\{c_t\}_{t=0}^{T}} E \sum_{t=0}^{T} \beta^t \left((1 - d_t) \frac{c_t^{1-\sigma}}{1-\sigma} + d_t \phi(a_{t-1}) \right)$$

$$c_t + a_t = y_t + (1 + r) a_{t-1} + b_t, \quad a_t \geq \underline{a}$$

1. **Bequests (or taste for wealth)**: $d_t = 1$ at death, warm glow $\phi(a)$, or $\kappa \frac{a^{1-\nu}}{1-\nu}$ with $\nu < \sigma$
   - wealth is a luxury good: $\bar{c} \to 0$ at the top, so $G_{\text{top}} \to 1 + r$
   - inheritances b_t also weaken the reset at death

Explaining wealth inequality

$$\max_{\{c_t\}_{t=0}^{T}} E \sum_{t=0}^{T} {\beta_i}^t (1 - d_t) \frac{c_t^{1-\sigma_i}}{1-\sigma_i}$$

$$c_t + a_t = y_t + (1+r)a_{t-1}, \quad a_t \geq \underline{a}$$

2. **Heterogeneous preferences**: patient households have a lower $\bar{c}$
   - random β makes saving rates random (route 2)
   - often chosen residually to match wealth (lecture 4, HKS 2021)

Explaining wealth inequality

$$\max_{\{c_t\}_{t=0}^{T}} E \sum_{t=0}^{T} \beta^t \left(1 - d_t\right) \frac{c_t^{1-\sigma}}{1-\sigma}$$

$$c_t + a_t = \left[l_e f\left(\theta_t, k_{t-1}\right) + \left(1 - l_e\right) y_t\right] + \left(1 + r\right)\left(a_{t-1} - k_{t-1}\right), \quad a_t \geq \underline{a}$$

3. **Entrepreneurship**: high, risky, persistent returns on own wealth, $k \leq \lambda a$
   - reduced form: "superstar" earnings state (Castañeda et al., 2003), which works through route 1

Explaining wealth inequality

$$\max_{\{c_t\}_{t=0}^{T}} E \sum_{t=0}^{T} \beta^t \left(1 - d_t\right) \frac{c_t^{1-\sigma}}{1-\sigma}$$

$$c_t + a_t = y_t + \left(1 + r_t^i\right) a_{t-1}, \quad a_t \geq \underline{a}$$

4. **Idiosyncratic rates of returns**: random growth (route 2) directly

Return heterogeneity: type vs scale dependence

- **Type dependence:** $r(z_{it})$, the return depends on a persistent type
  - capital income is linear in wealth within a type
  - $\zeta_{ra} = \zeta_a$ far in the tail
- **Scale dependence:** $r\,S(a)$ with $S(a) = 1 + \psi a^{\eta}$
  - the rich grow faster because they are rich
  - capital income $\propto a^{1+\eta}$, so $\zeta_{ra} = \zeta_a/(1+\eta)$
- Returns rise with wealth in the data, but richer households may also be higher types
  - panel data on returns help separate the two (Fagereng, Guiso, Malacrino and Pistaferri, 2020)

Explaining wealth inequality

Gaillard, Hellwig, Wangner and Werquin (2024)

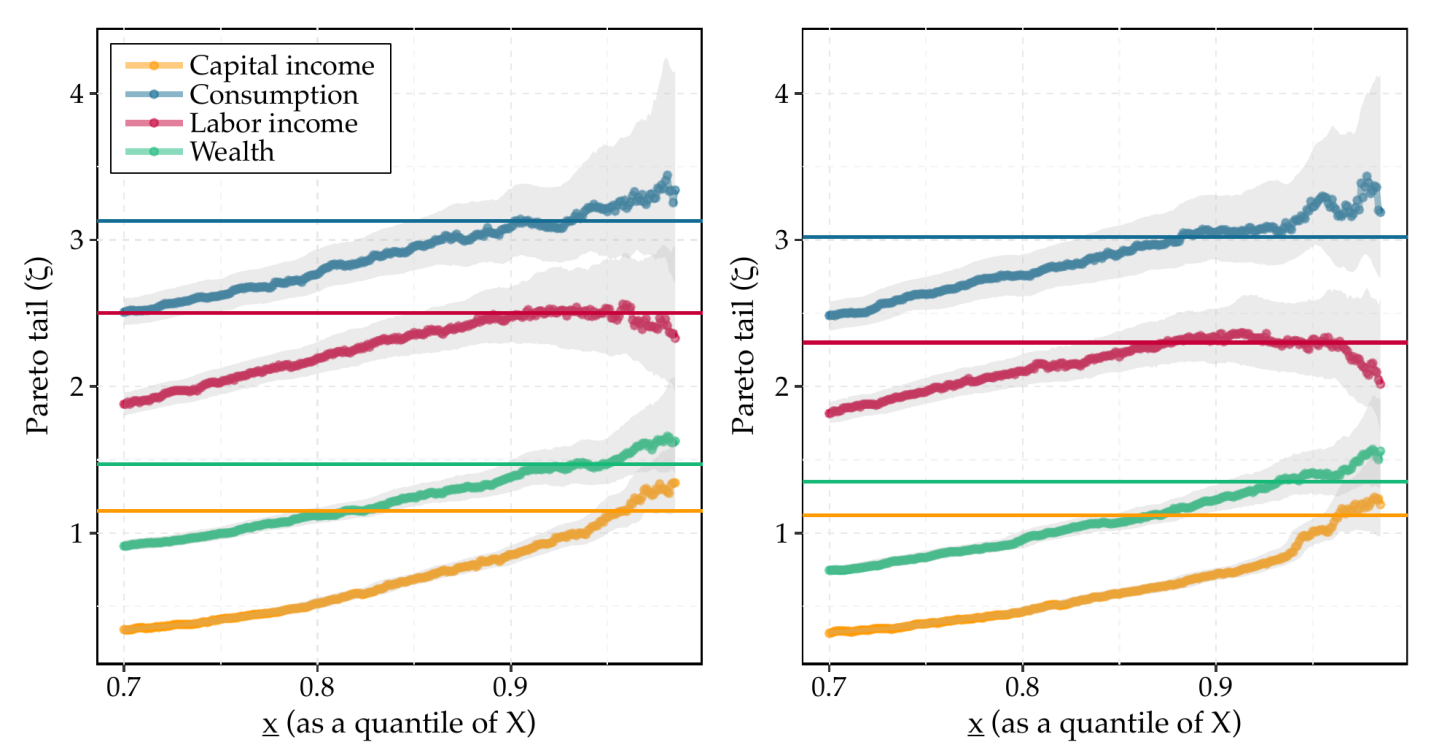
Gaillard et al. (2024)

- **Question:** many models generate high wealth inequality. Which ones match the ranking

$$\zeta_{ra} < \zeta_a < \zeta_y < \zeta_c \qquad (1.20,\ 1.38,\ 2.25,\ 3.06)$$

- **Answer:** two ingredients
  1. non-homothetic preferences: decouple consumption from wealth
  2. scale-dependent returns: decouple capital income from wealth

Empirical result: a strict ranking of Pareto tails



- Each dot: $\hat{\zeta}$ estimated on households above the threshold $\underline{x}$ (a quantile); lines: preferred estimates. PSID 2005–2021, raw (left) and adjusted for under-reporting (right)
- The ranking $\zeta_c > \zeta_y > \zeta_a > \zeta_{ra}$ holds at every threshold

Figure 4 from Gaillard et al. (2024)

Model

$$V(y,z,a) = \max_{c,\, a' \geq 0} \frac{c^{1-\gamma}}{1-\gamma} + \kappa \frac{(a/A)^{1-\nu}}{1-\nu} + \beta(1-\xi)E\left[V(y',z',a')\right]$$

$$\text{s.t.} \quad c + a' = wy - T(wy) + (1-\tau_K) r z S(a) a + a$$

- **Taste for wealth**, $\nu < \gamma$: wealth is a luxury good
- **Return type** $z \in \{1, z_H\}$, worker or entrepreneur: enter with probability 0.02, exit with 0.2
- **Scale dependence** $S(a) = 1 + \psi a^{\eta}$: the return per unit of wealth rises with wealth
- Labor productivity: AR(1), with a Pareto tail ($\zeta_h = 2.25$) at the top; HSV labor tax
- Death $\xi = 0.02$, bequests fully taxed: death resets wealth
- Cobb-Douglas firms; capital is effective wealth, $K = \int zS(a)a\, dF$

Intuition: two ingredients, two tails

- Power rule: if $X \propto a^b$ at the top, then $\zeta_X = \zeta_a / b$
  - concave in wealth: thinner tail; convex: fatter tail
- **Consumption** needs to be concave in wealth to have $\zeta_c > \zeta_a$
  - homothetic preferences: the rich consume a constant share of wealth, $c \propto a$, so $\zeta_c = \zeta_a$
  - taste for wealth: the rich save most of their income, c is concave in a, so $\zeta_c > \zeta_a$
- **Capital income** needs to be convex in wealth to have $\zeta_{ra} < \zeta_a$
  - same return for all: capital income $\propto a$, so $\zeta_{ra} = \zeta_a$
  - scale dependence: capital income $\propto a^{1+\eta}$, so $\zeta_{ra} = \zeta_a / (1 + \eta)$

Calibration

- **Fixed:** $\beta = 0.90$, $\gamma = 2$, $\xi = 0.02$, $\alpha = 0.33$, $\delta = 0.08$, $\tau_L = 0.14$, $\tau_K = 0.15$
- **Estimated (SMM):**

κ	ν	z_H	η	ψ
0.14	0.76	4.10	0.24	0.15

- **Targets:** the ratios of Pareto tails (ζ_{ra}/ζ_a, ζ_c/ζ_a, ζ_a/ζ_y), $K/Y = 3.8$, top 1% wealth share
- Tails in the model are estimated as in the data, on simulated PSID-like samples

Results

	ζ_c	ζ_a	ζ_{ra}	Top 1%
Data (adjusted PSID)	3.06	1.38	1.20	35%
(1) Homothetic, homogeneous returns	3.04	2.42	2.42	9%
(2) Homothetic, type dependence	2.65	1.32	1.02	29%
(3) Homothetic, scale dependence	2.56	1.30	1.16	32%
(4) Homothetic, type and scale	2.65	1.34	1.08	35%
(5) Non-homothetic, type dependence	3.08	1.37	1.19	34%
(6) Non-homothetic, type and scale	3.06	1.36	1.18	35%

Table 4 from Gaillard et al. (2024), each model re-estimated

What we learn

- **(1) Aiyagari with Pareto earnings:** wealth inherits the labor income tail (route 1). Far too thin: the top 1% hold 9%
- **(2) Entrepreneurs (type dependence):** high returns for some create random growth (route 2), which fixes the wealth tail
  - but the rich consume a constant share of their wealth: consumption is as unequal as wealth at the top
- **(3)-(4) Scale dependence:** also fixes wealth, but makes consumption even more unequal: the richest earn higher returns and consume them
- **(5)-(6) Taste for wealth:** the rich save most of their income, so consumption is much less unequal than wealth

What we learn (continued)

- Many models match wealth inequality; the **other tails** discipline them
- The consumption tail says: the rich save because they value wealth (or bequests), not only because their returns are high
- Type and scale dependence look the same in the PSID: only the very top separates them. Within the top 1%:
  - without scale dependence (5): $\zeta_{ra} = \zeta_a = 1.56$
  - with scale dependence (6): $\zeta_{ra} = 1.39 < \zeta_a = 1.47$
  - Saez and Zucman data: $1.28 < 1.55$

Explaining wealth inequality

Guvenen et al. (2023)

Wealth taxation

- Guvenen et al. (2023): *Use It or Lose It: Efficiency and Redistributional Effects of Wealth Taxation*
- Without return heterogeneity, a wealth tax and a capital income tax are equivalent:

$$a + (1 - \tau_k)ra = (1 - \tau_w)a + ra \quad \text{if} \quad \tau_w = r\tau_k$$

- With heterogeneous returns r_i:
  - capital income tax: falls on productive (high r_i) households, distortionary
  - wealth tax: same tax for the same wealth, shifts the burden to unproductive households

Model

$$a_t + c_t = \mathcal{W}(a_{t-1}, z_t) + w_t e_t \ell_t$$

$$\mathcal{W}(a_{t-1}, z_t) = \begin{cases} a_{t-1} + (\pi(a_{t-1}, z_t) + r a_{t-1})(1 - \tau_k) & \text{capital tax} \\ a_{t-1}(1 - \tau_w) + \pi(a_{t-1}, z_t) + r a_{t-1} & \text{wealth tax} \end{cases}$$

- Entrepreneurial profits: $\pi(a, z) = \max_{k \leq \lambda a} \{pzk - (r + \delta)k\}$
- Ability $z \in \{0, z_L, z_H\}$: workers, unproductive and productive entrepreneurs
- Warm-glow bequests, matches the Pareto tail of US wealth

Results

Replace $\tau_k = 25\%$ by a revenue-neutral wealth tax ($\tau_w = 1.2\%$)

TABLE V

TAX REFORM: CHANGE IN MACRO VARIABLES FROM CURRENT U.S. BENCHMARK

	Quantities (% change)						Prices (change)			
	K	Q	TFP_Q	L	Y	C	$\overline{w}$	$\overline{w}$ (net)	$\Delta r^\dagger$	$\Delta r^\dagger$ (net)
RN reform	16.4	22.6	5.3	1.2	9.2	9.5	8.0	8.0	0.21	–0.36
BB reform	9.2	16.0	6.2	1.2	6.9	7.7	5.6	5.6	0.67	–0.38

- Capital, productivity, output, consumption and wages increase
- Welfare gain of about 7% in consumption-equivalent terms

Optimal taxation

OPTIMAL TAXATION: TAX RATES AND AVERAGE WELFARE EFFECTS

	Benchmark U.S. economy	RN reform	OWT	OWT L-INEQ	OWT-X	WTE-X	OKIT
		(1)	(2)	(3)	(4)	(5)	(6)
Tax rates							
τ_k	25.0	—	—	—	—	—	–13.6
τ_a	—	1.19	3.03	2.54	3.80†	3.30	—
τ_ℓ	22.4	22.4	15.4	18.1	14.4	17.7	31.2
ΔWelfare							
$\overline{CE}_1$	—	6.8	9.0	6.0	9.1	8.4	4.2
$\overline{CE}_2$	—	7.2	8.7	5.2	8.8	8.6	5.1

- Optimal wealth tax: $\tau_w = 3.03\%$, welfare gain of 9%
- Optimal capital income tax: a subsidy of 13.6%, welfare gain of 4.2%
- $\rightarrow$ the source of inequality matters for policy

Drivers

Gabaix, Lasry, Lions and Moll (2016)

- Top income inequality in the US has risen fast since 1980
- Standard random growth models explain the Pareto tail, but their transitions are too slow
- Two fixes:
  - scale dependence, e.g. changes in skill prices
  - type dependence: "high-growth types", superstar entrepreneurs or managers
- Both are ingredients of Gaillard et al. (2024) for wealth

Drivers

Hubmer, Krusell and Smith (2021)

Hubmer, Krusell and Smith (2021)

- *Sources of US wealth inequality: past, present, and future*
- **Question:** can the observed changes in taxes, earnings and asset returns since 1967 explain the rise in US wealth inequality?
- **Method:**
  1. an Aiyagari model with return heterogeneity, calibrated so that 1967 is a steady state
  2. feed in the measured changes, 1967 to 2012
  3. compare the path of top wealth shares with the data
  4. counterfactuals: one change at a time

Household problem

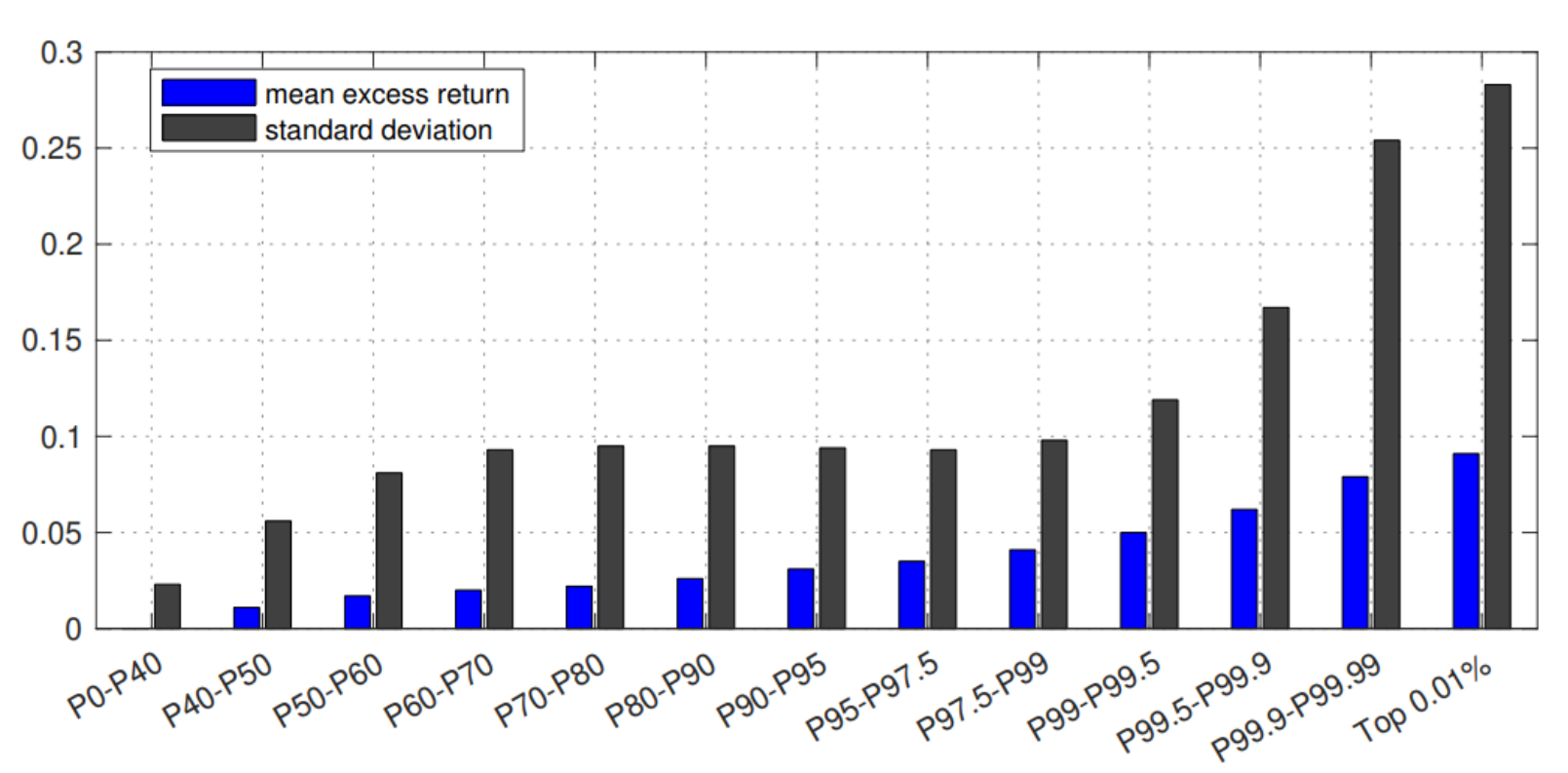
$$
\begin{aligned}
v_t(\omega_t, a_{t-1}) = \max_{a_t \geq \underline{a}} \frac{c_t^{1-\gamma}}{1-\gamma} + \beta_t E_t \left[v_{t+1}(\omega_{t+1}, a_t) \right] \\
\text{s.t.} \quad c_t + a_t = a_{t-1} + y_t - \tau_t(y_t) + (1 - \tilde{\tau}_t)\tilde{y}_t + T_t \\
y_t = \left(r_t + r_t^X(a_{t-1}) \right) a_{t-1} + w_t z_t \\
\tilde{y}_t = \sigma^X(a_{t-1}) \eta_t a_{t-1}
\end{aligned}
$$

- y_t: labor income and expected capital income, taxed progressively by $\tau_t(\cdot)$
- $\tilde{y}_t$: random capital income, $\eta_t \sim \mathcal{N}(0,1)$ i.i.d., taxed at a flat rate $\tilde{\tau}_t$
- **Returns** $r_t^X(a)$, $\sigma^X(a)$ rise with wealth (Swedish portfolio data)
- State $\omega_t = (z_t, \beta_t, \eta_t)$: z_t has persistent and transitory parts and a Pareto tail; β_t is AR(1)

Equilibrium: two market-clearing conditions

- Two aggregate unknowns, K_t and r_t, and two conditions:
  1. **capital**, as usual:
  $$K_t = \int a \, d\Gamma_t$$
  2. **capital income**: what households earn equals what firms pay
  $$\big(F_K(K_t, L) - \delta\big) K_t = \int \big(r_t + r_t^X(a)\big) a \, d\Gamma_t$$
- The schedule $r_t^X(\cdot)$ is exogenous: only its shape matters, r_t adjusts its level
- Without excess returns ($r^X = 0$), condition 2 is $r_t = F_K - \delta$: the usual Aiyagari model
- Return shocks $\sigma^X(a)\eta$ have mean zero and drop out; government: $T_t = 60\%$ of tax revenue

Schedule of excess returns



Data sources: Bach, Calvet, Sodini (2019); Kartashova (2014); Jorda et al. (2019); Case-Shiller.

Steady state fit (1967)

	Top 10%	Top 1%	Top 0.1%	Top 0.01%
Data	70.8	27.8	9.4	3.1
Single-β model	66.6	23.7	11.2	7.2
Benchmark model	73.8	27.4	8.4	3.2

- β heterogeneity chosen residually to match the 1967 distribution
- Return heterogeneity is the key ingredient for the top
- Progressive taxes and labor income risk dampen concentration

What drives steady-state inequality?

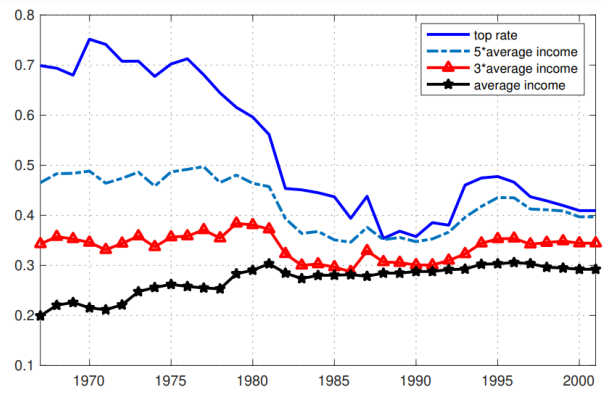
Channel	Top 10%	Top 1%	Top 0.1%	Top 0.01%	Gini
β heterogeneity	8.8	7.7	3.8	2.0	0.050
Earnings heterogeneity	−27.5	−17.8	−9.5	−6.4	−0.173
persistent	−5.0	−7.5	−4.2	−2.9	0.009
transitory	−11.6	−4.3	−1.7	−0.9	−0.109
Tax progressivity	−21.3	−61.8	−71.2	−67.1	−0.148
Return heterogeneity	29.5	18.4	6.6	2.8	0.192
mean differences	25.8	16.7	6.0	2.6	0.174
return risk	0.7	2.2	3.3	2.5	0.004

- Benchmark minus the model without the channel, in pp of wealth
- **Tax progressivity:** with a flat tax, the top 1% share goes from 27% to 89%
  - rich households save $\approx \left(\beta(1 + r(1 - \tau'(y)))\right)^{1/\gamma}$: progressivity lowers G_{top}
- **Earnings risk reduces** top shares: more precautionary saving at the bottom, and a lower r

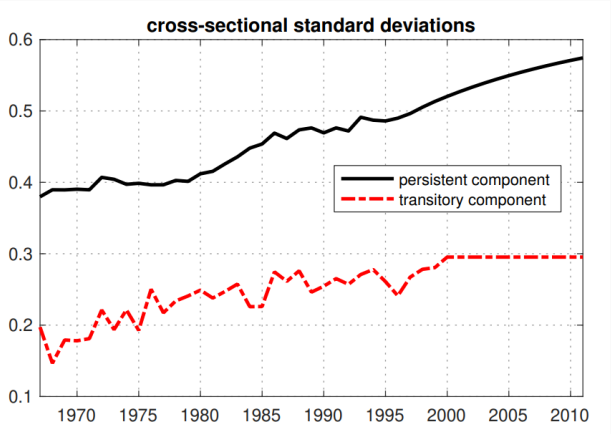
Main experiment: the 1967–2012 transition

- Start in the 1967 steady state. Only the 1967 distribution is targeted
- In 1967, households learn the whole future path of four exogenous changes:
  1. **taxes** $\tau_t(\cdot)$, $\tilde{\tau}_t$: progressivity falls from the late 1970s
  2. **earnings risk**: the variances of the shocks to z_t rise (Heathcote, Perri and Violante, 2010)
  3. **top earnings**: the Pareto tail of z_t gets fatter (Piketty and Saez)
  4. **excess returns** by asset class: 10-year moving averages
- Fixed: preferences, portfolio weights by wealth, return dispersion
- Perfect-foresight transition: K_t, r_t, T_t clear markets every year
  - same results with myopic households who expect today's environment to last forever

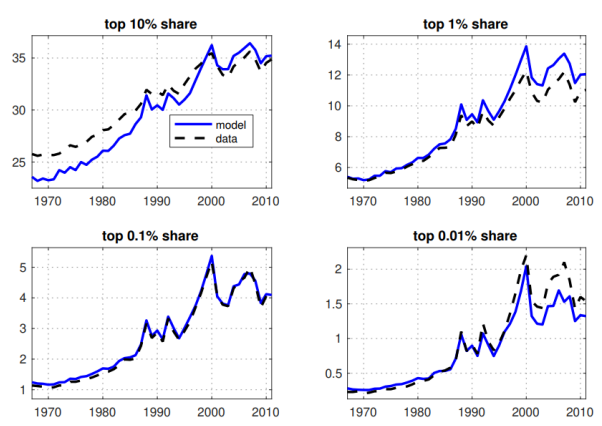
Transition: four observed changes, 1967 to 2012



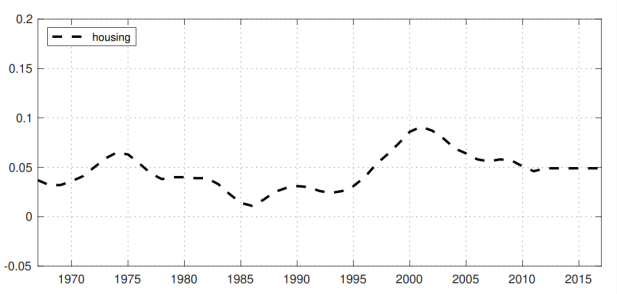
(a) Tax progressivity



(b) Labor income risk

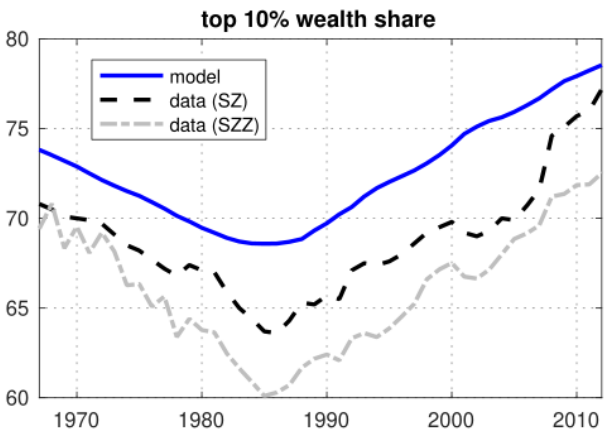


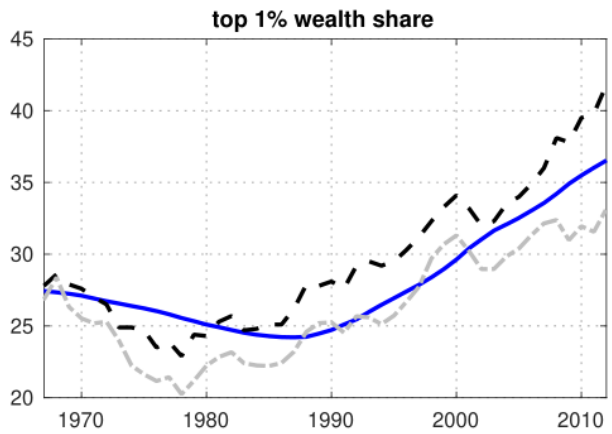
(c) Top labor income shares

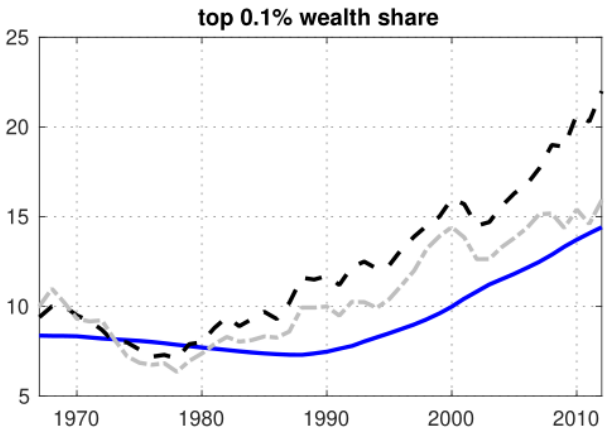


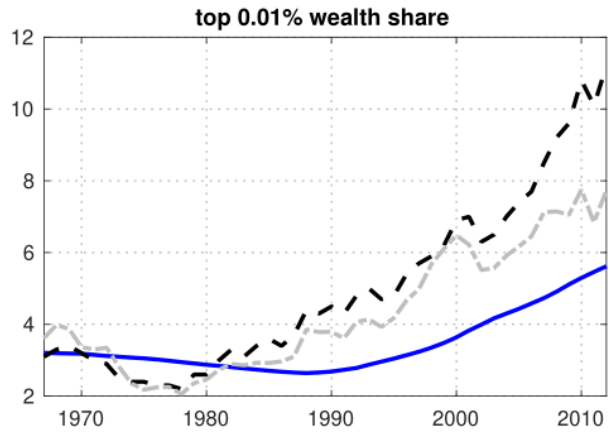
(d) Return premia

Historical evolution



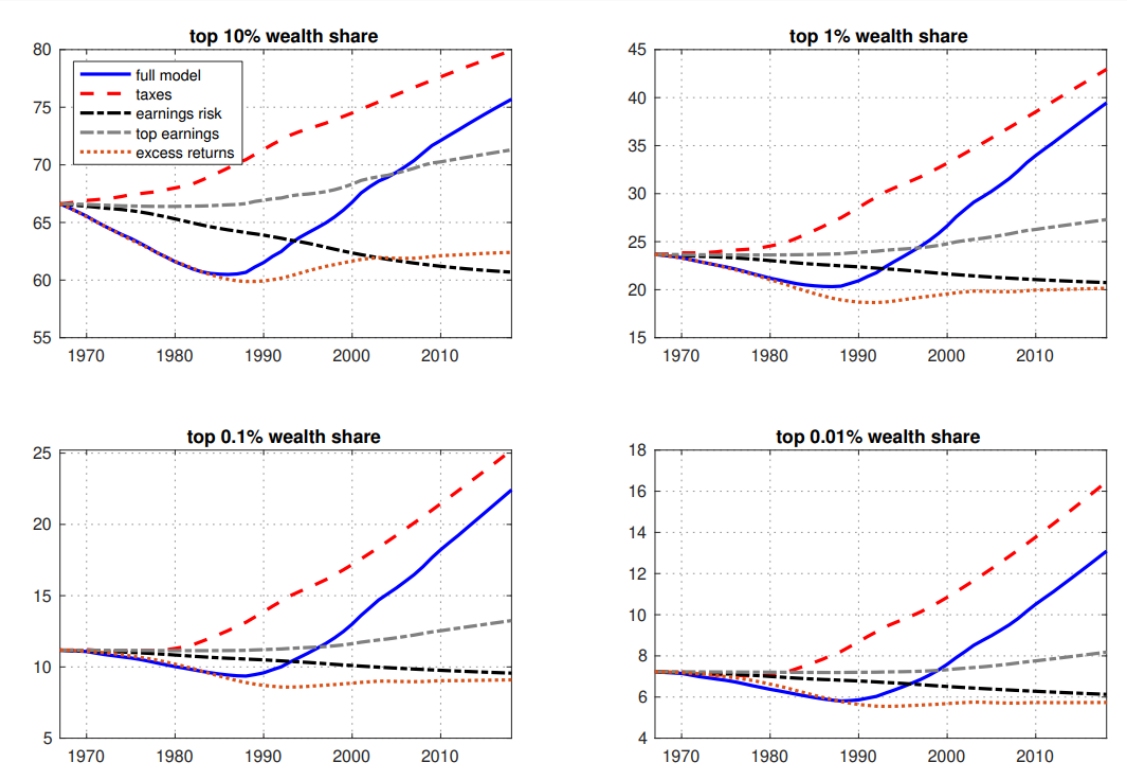






Data sources: dashed black lines refer to Saez & Zucman (2016); dash-dotted gray lines refer to Smith et al. (2020).

Decomposition



- Lower tax progressivity alone more than explains the rise of the top 10% and top 1% shares (157% and 115%), less at the very top
- Return premia explain the U-shape
- Higher earnings risk lowers wealth inequality: more precautionary saving at the bottom

Hubmer, Krusell and Smith: what we learn

- Return heterogeneity, across and within wealth groups, is enough to match 1967 wealth inequality and its Pareto tail
  - β heterogeneity only plays a small, residual role
- Tax progressivity is the main brake on wealth concentration
  - its fall since the late 1970s is the main driver of the rise
- Swings in asset returns explain the U-shape: low stock and private equity returns in the 1970s, then a recovery
- Higher earnings inequality does not explain the rise: more earnings risk even lowers top shares
- The transition is far from over: with today's taxes, top shares keep rising for decades (slow random growth, as in Gabaix et al.)
- Limit: the model explains only about 30% of the rise of the top 0.01%

Drivers

Moll, Rachel and Restrepo (2022)

Moll, Rachel and Restrepo (2022): model

Households with skill z (wage percentile, population share ℓ_z), continuous time:

$$\max \int_0^\infty e^{-\rho t} \frac{c_t^{1-\sigma}}{1-\sigma} dt \quad \text{s.t.} \quad \dot{a}_t = w_z + r a_t - c_t$$

- same return r for everyone, no income risk
- dissipation shock at rate ξ: wealth reset to 0 (death without bequest, failed business, ...); ρ includes ξ

Firms: tasks done by labor or capital; a share α_z of the tasks of skill z is automated

$$Y = AK^\alpha \prod_z (\psi_z \ell_z)^{(1-\alpha_z)\eta_z}, \qquad \alpha = \sum_z \alpha_z \eta_z$$

- factor prices: $w_z \ell_z = (1-\alpha_z)\eta_z Y$ and $r + \delta = \alpha Y/K$
- **automation**: α_z rises, tasks move from skill z to capital

Equilibrium: K = household wealth, summed over skills and over time since the last reset

Key result: top inequality = net capital share

- Saving is linear in total wealth $x = a + w/r$, which grows at $g = \frac{r-\rho}{\sigma}$: the constant $\bar{G}$ + reset model, so

$$\frac{1}{\zeta} = \frac{g}{\xi}$$

- Steady state: wealth added by savers = wealth lost to resets, with $\bar{w} = \sum_z w_z \ell_z$ the wage bill

$$g\left(K + \frac{\bar{w}}{r}\right) = \xi K \quad \Rightarrow \quad \frac{1}{\zeta} = \frac{g}{\xi} = \frac{rK}{rK + \bar{w}} = \alpha_{net}$$

- **Top inequality = net capital share**
  - all capital is rebuilt from savings: the more capital the economy needs, the faster savers must grow, the fatter the tail

Application: automation

- Automation shifts tasks from labor to capital: more demand for capital
  - r and α_{net} rise: fortunes grow faster, fatter tail
  - part of the productivity gain goes to capital: wages can fall
- Experiment: automation of routine jobs, 1980–2014, calibrated to the rise of the capital share from 34.5% to 42.8%
- Results (steady states of the extended model, with risky capital and growth, so $1/\zeta$ is no longer exactly α_{net}):
  - top income tail $1/\zeta$ from 0.54 to 0.65, about 70% of the observed rise
  - wages fall below the 80th percentile; incomes $+20$ to $+55$% within the top 1%, mostly capital income
- Caveat: like all random growth models, the transition is too slow (Gabaix et al.)

Exercise

Exercise: HANC model with return heterogeneity

$$v_t(e_{it}, r^X_{it}, a_{it-1}) = \max_{c_{it}, a_{it}} \frac{c_{it}^{1-\sigma}}{1-\sigma} + \beta E_t \left[v_{t+1}(e_{it+1}, r^X_{it+1}, a_{it}) \right]$$

$$\text{s.t.} \quad a_{it} + c_{it} = (1 + r_t + r^X_{it}) a_{it-1} + w_t e_{it}, \qquad a_{it} \geq 0$$

$$\log e_{it} = \rho_e \log e_{it-1} + \psi_{it}, \qquad \psi_{it} \sim \mathcal{N}(\mu_\psi, \sigma^2_\psi)$$

$$r^X_{it} = (1 - \rho_r)\bar{r}^X + \rho_r r^X_{it-1} + \epsilon_{it}, \quad \epsilon_{it} \sim \mathcal{N}(0, \sigma^2_r)$$

- **Q1:** solve the household problem in partial equilibrium. What are average returns? Are they equal to $r + \bar{r}^X$?
- **Q2:** why do we need a new market-clearing condition? Write it down
- **Q3:** calibrate the HANC model with return heterogeneity in general equilibrium
- **Q4:** calibrate a standard HANC model without return heterogeneity and compare the two wealth distributions

Summary

Summary and next week

- **Today:** wealth inequality
  1. Pareto tails come from random growth, stopped by a reset
  2. Buffer-stock model fails: the rich dissave
  3. Fixes: taste for wealth, return heterogeneity, entrepreneurs, bequests
  4. Rise since 1980: taxes, return premia, automation
- **Homework exercise:** HANC model with return heterogeneity
  - see `07-Wealth-Inequality/exercise-het-return` on GitHub
- **Next week:** Secular stagnation

Appendix: why $E[G^\zeta] = 1$

Guess and verify: guess $P(a > x) \approx Cx^{-\zeta}$ for large x

1. **Who is above x tomorrow?** For large x, y is negligible, so $a_{it} > x$ iff $a_{it-1} > x/G_{it}$
2. **Pareto scaling:** with the guess, the number above x/G is G^ζ times the number above x (only a power law scales this way)

$$P\left(a > x/G\right) \approx C\left(\frac{x}{G}\right)^{-\zeta} = G^\zeta\, Cx^{-\zeta}$$

3. **Average over G** (independent of a_{it-1}):

$$P(a_{it} > x) \approx E\left[G^\zeta\right]\, P(a_{it-1} > x)$$

   $E[G^\zeta]$: factor by which the number of households above x changes in one period

4. **Stationarity:** this number must not change, at any large x:

$$\boxed{E\left[G^\zeta\right] = 1}$$

Lognormal G, $\ln G \sim \mathcal{N}(\mu_G, \sigma_G^2)$: $E[G^\zeta] = e^{\zeta\mu_G + \zeta^2\sigma_G^2/2} = 1$, so $\zeta = -2\mu_G/\sigma_G^2$

◂ Back

Appendix: a counting example

- Each year, half of the rich grow by 20% ($G = 1.2$), half shrink by 30% ($G = 0.7$). N households are above \$1M today
- Above \$1M next year:
  - lucky half: those above \$833k today, $\frac{1}{2}1.2^{\zeta}N$
  - unlucky half: those above \$1.43M today, $\frac{1}{2}0.7^{\zeta}N$
- Stationarity: $\frac{1}{2}1.2^{\zeta} + \frac{1}{2}0.7^{\zeta} = 1$

ζ	1.2^{ζ}	0.7^{ζ}	$E[G^{\zeta}]$
2	1.44	0.49	0.97: tail shrinks
3	1.73	0.34	1.04: tail grows
2.6	1.61	0.40	1.00: steady state

Same counting at \$10M or \$1B: one ζ for the whole tail

Back